

Quantifying the Effective Sample Size of Task Descriptors in In-Context Learning

Bruce Quan Nguyen

October 1, 2026

Abstract

In-context learning (ICL) allows models to adapt to tasks using prompts that typically combine a natural language task descriptor with input-output demonstrations. While the informational value of demonstrations is well understood, quantifying the precise contribution of the task descriptor remains challenging. By formulating ICL as Bayesian inference over a latent task, we treat the descriptor and demonstrations as distinct modalities of observations. We quantify the descriptor’s value via its *effective sample size* (ESS)—the equivalent number of demonstrations required to achieve an identical reduction in one-step predictive log-loss regret for a Bayes-optimal predictor. In the exact setting of in-context linear regression, we demonstrate that a descriptor’s ESS is governed by its *precision* (the reduction in prior variance) and its *reliability* (the probability of correct specification). We prove that for a correctly specified descriptor, ESS is about $(d - 1)(1 - r) + 1/(ra_0)$, where r is the fraction of prior variance the descriptor leaves and a_0 the prior signal-to-noise ratio: it scales linearly with task dimension d and with precision $1/r$. Conversely, for a potentially misspecified descriptor, the ESS is strictly bounded without demonstrations, as the optimal predictor must maintain posterior uncertainty over the descriptor’s validity. Such a descriptor retains at least a fraction $1 - p$ of the regret of predicting from the prior alone: at $p = 0.9$ in sixteen dimensions, a hundredfold gain in precision adds ten demonstrations, against a hundred when correctly specified. Demonstrations act as evidence to resolve this unidentifiability, effectively unlocking the descriptor’s value. The ESS then approaches p times that of a correctly specified descriptor; the regime that matters is a precise but imperfect descriptor with few demonstrations. Furthermore, we show that small meta-trained Transformers behave Bayes-optimally within the constraints of their output representations; when faced with misspecified descriptors, unimodal output heads force the model to project the optimal bimodal mixture onto a single Gaussian. They reproduce the ESS of a correctly specified descriptor, in one training run per setting. Finally, we demonstrate that a network’s reliance on the descriptor is entirely determined by its meta-training distribution, where over-reliance on highly precise but misspecified descriptors yields catastrophic regret. Our claims concern Bayes-optimal predictors and small meta-trained models, not large language models.

1 Introduction

The empirical success of large language models is heavily reliant on in-context learning (ICL), where a model adapts to a novel task given a prompt (Brown et al., 2020). Standard prompts incorporate two distinct sources of task information: a natural language task descriptor and a few demonstration examples. Empirical observations regarding the descriptor’s utility are heavily mixed; while a well-crafted descriptor can substitute for hundreds of examples (Le Scao and Rush, 2021; Honda et al., 2025), models are also known to perform adequately with misleading descriptors (Webson and Pavlick, 2022) or heavily prioritize demonstrations over explicit instructions (Gupta et al., 2025). Existing theoretical frameworks for ICL often model the prompt strictly as a sequence of demonstrations (Garg et al., 2022; Akyürek et al., 2023; Xie et al., 2022), leaving a fundamental theoretical gap: how can we rigorously quantify the informational equivalent of a task descriptor?

We address this by adopting a Bayesian perspective on ICL (Xie et al., 2022). In this framework, the pretraining distribution constitutes a prior over a latent task. The task descriptor acts as a direct, explicit observation of the task parameters, whereas demonstrations serve as indirect observations. Conditioning on the descriptor yields an updated task prior. This is the sense in which “the descriptor is the prior”: it is the prior under which the demonstrations are read. In Bayesian statistics, the information contained in a prior can be quantified as its effective sample size (ESS) (Morita et al., 2008), which is dictated by the prior’s concentration (its *precision*) and the probability of prior-data conflict (its *reliability*) (Evans and Moshonov, 2006).

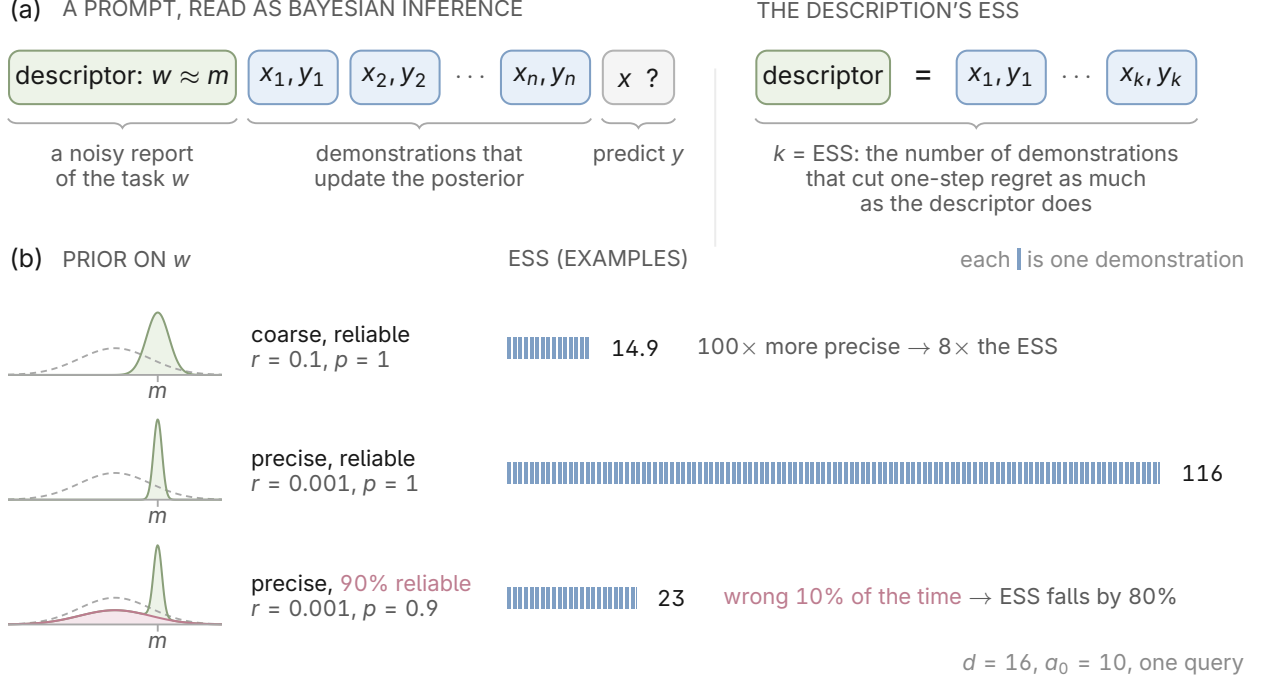


Figure 1: **A task descriptor is worth a number of demonstrations, set by its precision and capped by its reliability.** (a) We formulate a prompt as Bayesian inference over a latent task w under the pretraining prior: the descriptor is a direct noisy observation of w and the demonstrations are indirect ones. The descriptor’s effective sample size (ESS) is the number of demonstrations k that lowers a Bayes-optimal predictor’s one-step regret by as much as the descriptor does. (b) Three descriptors and their ESS. The precision r sets how narrowly the prior concentrates around the stated m ; the reliability p is the probability that the descriptor is correctly specified, and otherwise w follows the base prior (dashed; its share $1 - p$ shaded pink). Prior shapes are sketches; ESS values are from Table 3 in Appendix A ($d = 16, a_0 = 10$).

We analyze this dynamic within noisy in-context linear regression with a Gaussian task prior (Garg et al., 2022; Akyürek et al., 2023; Raventós et al., 2023), where Bayes-optimal inference admits closed-form solutions. This tractability allows us to exactly compute a descriptor’s ESS and provides exact reference predictors to evaluate the behavior of meta-trained Transformer models (Genewein et al., 2025). The reliability of the descriptors a network is meta-trained on is a parameter we control, so the reliance a Transformer learns can be read off against two exact references: the predictor calibrated to the reliability and the one that relies on the descriptor fully. A descriptor is a noisy observation of the regression weights, and a potentially misspecified descriptor follows the robust mixture prior that clinical trials use to admit possibly conflicting historical data (Schmidli et al., 2014).

Contributions.

- We define the ESS of a task descriptor by matching the one-step log-loss regret of a Bayes-optimal predictor given the descriptor to that given demonstrations alone, with or without further demonstrations already in the prompt. It is the prior sample size of Reimherr et al. (2021) with predictive regret as the measure of uncertainty (§3).
- We prove that a correctly specified descriptor has $\text{ESS} \approx (d - 1)(1 - r) + 1/(ra_0)$: the fraction of the dimension it constrains at high signal-to-noise ratio, plus the conjugate-prior ESS of its own prior. We establish the expansion in the coarse and precise regimes, with error terms, and a bound that holds everywhere (§4).
- We prove that a potentially misspecified descriptor’s regret is at least $1 - p$ times that of predicting from the prior alone, and at most that plus an identification cost of $H(p)$, so without demonstrations its ESS saturates however precise it is. Demonstrations resolve this unidentifiability and lift the cap: with n demonstrations in hand the

ESS approaches p times that of a correctly specified descriptor. An overconfident predictor is instead harmed by precision (§5).

- We meta-train small Transformers on the same generative process and place each between exact reference predictors. They are Bayes-optimal within their output class: they reproduce the ESS of a correctly specified descriptor and, with a potentially misspecified descriptor alone, match the best single Gaussian rather than the Bayes-optimal mixture their head cannot express, converging with it as demonstrations arrive. Their reliance is inherited from meta-training: trained only on correctly specified descriptors they act as overconfident predictors, and trained on potentially misspecified ones they keep hedging when every descriptor is correct (§6).

Together these results establish that reliability is not a fixed discount on a descriptor’s value but a cost that demonstrations pay down.

2 Related Work

Our approach builds upon the understanding of ICL as implicit Bayesian inference (Xie et al., 2022), where meta-trained Transformers have been shown to implement optimal estimators such as Bayes-optimal ridge regression (Akyürek et al., 2023; Raventós et al., 2023; Panwar et al., 2024). While some literature has integrated task descriptors—modeling them as tokens that indicate hypothesis classes or carry input means—these works generally do not quantify the descriptor’s value in units of examples, nor do they formalize the probability of descriptor misspecification (Huang and Ge, 2025; Lin et al., 2025; Tong et al., 2026; Lin and Lee, 2024). Our ESS formulation aligns with prior sample size methodologies in Bayesian statistics, particularly those utilizing predictive regret (Morita et al., 2008; Reimherr et al., 2021; Neuenschwander et al., 2020; Reznik, 2026). To model potential misspecification, we employ robust mixture priors, a technique commonly used in clinical trials to integrate potentially conflicting historical data (Schmidli et al., 2014).

Two of these works supply the comparison we make. Genewein et al. (2025) score meta-trained predictors by their expected excess log loss over the oracle that knows the task, and compare that regret with the Bayes-optimal predictor’s; we adopt both the measure and the comparison. Zhu et al. (2026) show that a Transformer given a prefix of related datasets performs amortised hierarchical Bayesian prediction and matches the oracle across prior families, the closest demonstration that a prior-carrying prefix is used Bayes-optimally. In contrast to all of the above, we express the value of the conditioning information in units of demonstrations, and we allow it to be misspecified. Among the descriptor models, Huang and Ge (2025) add a descriptor token carrying the input mean; it helps their learner by centring the inputs but carries no information about the weights, so to a Bayes-optimal predictor its ESS is zero. Lin et al. (2025) guide models with hypothesis classes, and Tong et al. (2026) show that a prediction’s dependence on an instruction, correct or not, decays exponentially in the number of demonstrations. Lin and Lee (2024) analyse Gaussian-mixture task priors whose components are fixed by pretraining; ours has two components, one centred by the descriptor.

Among the prior sample size methodologies, Morita et al. (2008) define the ESS of a prior by matching its information to that of a baseline prior plus a sample. Reimherr et al. (2021) make this prior sample size a function of the data in hand, allow it to be negative for a prior that conflicts with the likelihood, and leave prediction-based measures of uncertainty open; our ESS is their prior sample size with one-step predictive regret as that measure. Neuenschwander et al. (2020) note that a mixture prior’s ESS is not the weighted average of its components’. Reznik (2026) defines a borrowed ESS under predictive log loss for a power prior, with a divergence between historical and current data in the role our $1 - p$ plays; it has no mixture and does not depend on the current data. The worked example of Schmidli et al. (2014) places weight 0.1 on a vague component, our $p = 0.9$, with our base prior in the vague component’s role. In that literature’s terms a misspecified descriptor is a prior-data conflict (Evans and Moshonov, 2006).

3 Problem Formulation

Tasks and Demonstrations. We assume a latent task represented by a weight vector $w \in \mathbb{R}^d$. A demonstration consists of an input $x \sim \mathcal{N}(0, I_d)$ and a label $y = w^\top x + \epsilon$, with observation noise $\epsilon \sim \mathcal{N}(0, \sigma_y^2)$. The base prior over tasks is isotropic Gaussian $w \sim \mathcal{N}(0, s_0^2 I_d)$. The prior signal-to-noise ratio (SNR) is defined as $a_0 = s_0^2 / \sigma_y^2$, which we set to $a_0 = 10$. We set $\sigma_y = 1$. This a_0 is above the values of 1 to 4 in the noisy settings of Garg et al. (2022); Raventós et al. (2023); Panwar et al. (2024) and within the range swept by Akyürek et al. (2023), so that a precise descriptor can be worth many demonstrations.

Task Descriptors. A descriptor is formalized as a vector $m \in \mathbb{R}^d$. Its generative process depends on a latent Bernoulli indicator $Z \sim \text{Bernoulli}(p)$, which dictates whether the descriptor is correctly specified ($Z = 1$) or misspecified ($Z = 0$). The generative process is:

$$m \sim \mathcal{N}(0, (1-r)s_0^2 I_d) \quad (1)$$

$$w \mid m, Z = 1 \sim \mathcal{N}(m, rs_0^2 I_d) \quad (2)$$

$$w \mid m, Z = 0 \sim \mathcal{N}(0, s_0^2 I_d) \quad (3)$$

Marginalizing over Z , the prior for w given the descriptor m is the robust mixture $p\mathcal{N}(m, rs_0^2 I_d) + (1-p)\mathcal{N}(0, s_0^2 I_d)$ (Schmidli et al., 2014). The marginal law of w is the base prior in both cases, and the base prior is the mixture’s vague component. The parameter $r \in (0, 1)$ denotes the descriptor’s **precision** (the ratio of the conditional variance to the base prior variance). A smaller r is thus a more precise descriptor: it locates each weight to within $\sqrt{r}$ of its typical size s_0 , that is, 32% at $r = 0.1$, 10% at $r = 0.01$, and 3% at $r = 0.001$. The parameter p denotes the descriptor’s **reliability**. We refer to a descriptor with $p = 1$ as *correctly specified* and to one with $p < 1$ as *potentially misspecified*; a realized descriptor with $Z = 0$ is *misspecified*.

Two features of this model matter below. First, a correctly specified descriptor is data about the task: equivalently, $m \mid w \sim \mathcal{N}((1-r)w, r(1-r)s_0^2 I_d)$, a direct noisy observation of w , so the ESS is an exchange rate between direct observations of w and indirect ones through x . Second, a misspecified descriptor is unrelated to the task rather than adversarial. A descriptor that pointed away from w by a fixed rule would constitute information about w to a learner that knew the rule, and would harm only one that did not. Throughout, the descriptor is a numerical observation, not text to be parsed.

Regret and ESS. We evaluate a predictor’s uncertainty given a prompt D (the descriptor, n examples, and a query x) via its one-step expected excess log-loss regret relative to an oracle that knows the true task w :

$$R = \mathbb{E} [-\log q(y \mid D) + \log \phi(y; w^\top x, \sigma_y^2)] \quad (4)$$

where ϕ is the Gaussian density. Here $q(y \mid D)$ is the predictor’s predictive density and the expectation is over tasks, descriptors, inputs, and labels; this is the per-step version of the regret of Genewein et al. (2025). We use log loss rather than the squared error of Garg et al. (2022) because it is a proper scoring rule and the loss under which the mixture bound of §5 holds. Let $R^{\text{desc}}(n)$ denote the Bayes-optimal regret with the descriptor and n examples, and $R^{\text{ex}}(n)$ the regret with n examples alone. We write $R_{p=1}^{\text{desc}}(n)$ for the former with a correctly specified descriptor. Reimherr et al. (2021) measure a prior π against a baseline π_b given data I by the number M of further observations that make the baseline as uncertain as the prior: $U_{\pi_b}(I + M) = U_\pi(I)$. We take the base prior as π_b , the descriptor-conditioned mixture prior as π , the n demonstrations as I , and regret as U .

Definition 1. The Effective Sample Size (ESS) of a descriptor, given n demonstrations, is $\text{ESS}_n = n^* - n$, where n^* is the root of $R^{\text{ex}}(n^*) = R^{\text{desc}}(n)$, calculated via linear interpolation. The standalone ESS is ESS_0 .

The interpolation is of R^{ex} , linearly between integers. ESS_n is the number of further demonstrations a demonstrations-only predictor requires to match the predictor that also has the descriptor; it is negative when the descriptor hurts. Because both sides are Bayes-optimal predictors with different conditioning sets, the ESS quantifies the information in the descriptor rather than the architecture of any particular learner.

Reference Predictors and Reliance. A learner applies an effective mixture weight q to the descriptor component. We call q its *reliance* on the descriptor, the weight its posterior gives the component in which the descriptor is correctly specified; a learner need not know the reliability p . An *oracle-calibrated* predictor uses $q = p$ (Bayes-optimal). An *overconfident* predictor assumes $q = 1$. It takes every descriptor as correctly specified, as a learner that has never encountered a misspecified one would. Both are Bayes-optimal under their own belief about reliability. We define a learner’s *implied reliance* as the value q that minimizes the mean squared error between its posterior-mean predictions and those of the corresponding reference model. The mean is over prompts; the implied reliance locates the learner between the two reference predictors.

4 The Value of Precision (Correctly Specified Descriptors)

When a descriptor has perfect reliability ($p = 1$), the posterior remains Gaussian. Given n demonstrations $X_n \in \mathbb{R}^{n \times d}$, the posterior covariance of w under a Gaussian prior with SNR a , that is, with variance $a\sigma_y^2$ per coordinate, is $\Sigma = \sigma_y^2(I/a + X_n^\top X_n)^{-1}$, which depends on neither the labels nor the prior mean. The regret is half the expected log of the remaining predictive variance, $R = \frac{1}{2}\mathbb{E} \log(1 + x^\top \Sigma x)$. A correctly specified descriptor is the case $n = 0$, $a = ra_0$; demonstrations alone have $a = a_0$.

Proposition 1. *For $p = 1$ and conjugate-prior ESS $c = 1/(ra_0) = \sigma_y^2/(rs_0^2)$:*

1. (Coarse descriptors) *If $c \leq 1$, then as $rd \rightarrow \infty$, $ESS = (d-1)(1-r) + (1-r)c + O(1/(rd))$.*
2. (Precise descriptors) *If $c \geq d$, then as $c/d \rightarrow \infty$, $ESS = c + d + 1 - 1/a_0 + O(d/c)$.*
3. *For every d, a_0, r , we have $ESS \leq c + d + 2$.*

The expansions in (1) and (2) hold uniformly in $c \leq 1$ and in $a_0 \geq 1$, respectively; Appendix B.1 states and proves the proposition in a fuller form, with the regret expansions behind (1) and (2).

Proof idea. A descriptor shrinks the prior variance by the factor r in all d directions. Demonstrations remove variance differently. At high SNR each of the first d demonstrations removes the variance along the one direction it probes, so $n < d$ demonstrations leave $d - n$ directions untouched; matching that leftover to the descriptor's gives $d - n \approx rd$, the high-SNR term $(d-1)(1-r)$. Once $n \geq d$ every direction has been probed and the leftover decays like $d/(n-d-1)$; matching it to the descriptor's d/c gives $n \approx c + d + 1$. The base prior has conjugate-prior ESS $1/a_0$ of its own, which both sides share; it is the $-1/a_0$ in (2) and, through $(1-r)c = c - 1/a_0$, the correction in (1).

The constant $c = \sigma_y^2/(rs_0^2)$ is the ESS of the descriptor's own prior $\mathcal{N}(m, rs_0^2 I_d)$ in the conjugate sense, noise variance over prior variance (Neuenschwander et al., 2020), and §5 shows it is also the limit of ESS_n as $n \rightarrow \infty$; this is why we call it the conjugate-prior ESS. Both asymptotic regimes demonstrate that the ESS of a correctly specified descriptor decomposes into a high-SNR term (representing the fraction of the dimension it constrains) and the conjugate-prior ESS c . Roughly, $ESS \approx (d-1)(1-r) + 1/(ra_0)$. The rule holds within about two demonstrations: on the grid of Table 2 (Appendix A) it is within 2.2 demonstrations of the exact ESS in every cell with $a_0 \geq 10$. The regimes meet at $c = 1$, where the descriptor constrains w as tightly as one label's noise does, and the ESS passes d near $c = d$. Concretely, a descriptor that removes nine tenths of the prior variance is worth nine tenths of the dimension plus one demonstration, 14.9 demonstrations at $d = 16$ and 57.7 at $d = 64$; one that leaves a thousandth of the variance is worth its conjugate-prior ESS of 100 plus $d + 1$, or 117 demonstrations at $d = 16$. Figure 2(a) traces the full curve at $d = 16$.

5 The Identifiability Bottleneck (Potentially Misspecified Descriptors)

When $p < 1$, the Bayes-optimal posterior predictive is a two-component mixture. Its components are one that relies on the descriptor and one that ignores it, weighted by the posterior probability that the descriptor is correctly specified. The optimal predictor must maintain uncertainty over the validity of the descriptor. The following proposition bounds what this hedge costs, with or without demonstrations to check the descriptor against.

Proposition 2. *Fix $n \geq 0$, $p \in (0, 1)$, and $r \in (0, 1)$. Let D_n be the prompt (the descriptor, n demonstrations, and the query) and $\pi_n = P(Z = 1 \mid D_n)$ the posterior probability that the descriptor is correctly specified. Let $\pi_n(Z)$ denote the posterior weight on the true mixture component. That is, $\pi_n(Z)$ is π_n if $Z = 1$ and $1 - \pi_n$ otherwise. The regret is bounded by:*

$$pR_{p=1}^{desc}(n) + (1-p)R^{ex}(n) \leq R^{desc}(n) \leq pR_{p=1}^{desc}(n) + (1-p)R^{ex}(n) + \mathbb{E}[-\log \pi_n(Z)] \quad (5)$$

The predictor does no better than an oracle twin who is told whether the descriptor is correctly specified, and no worse than that twin plus the cost of not knowing. The identification cost $\mathbb{E}[-\log \pi_n(Z)]$ equals the binary entropy $H(p)$, in nats, at $n = 0$ and is non-increasing in n . As $r \rightarrow 0$ the standalone regret lies between $(1-p)R^{ex}(0)$ and $(1-p)R^{ex}(0) + H(p)$.

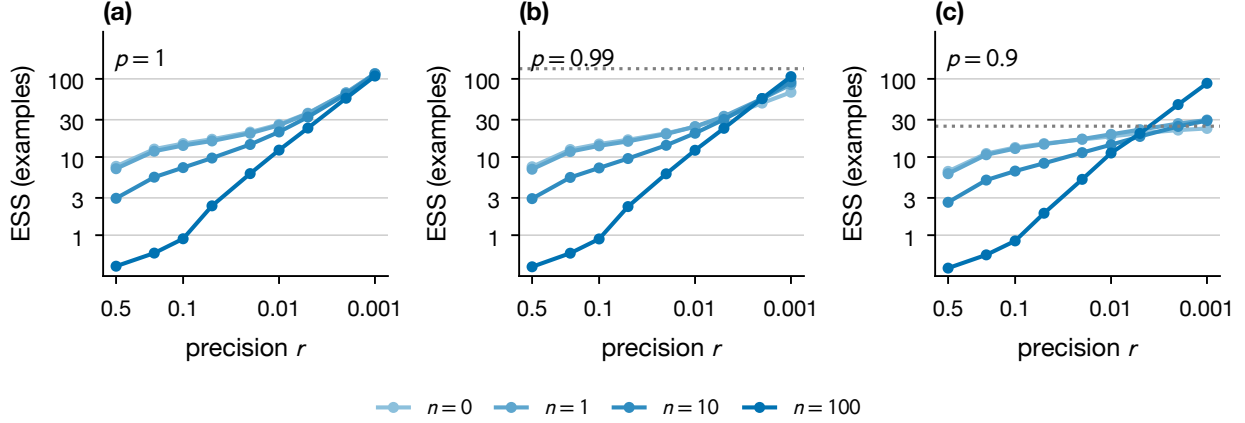


Figure 2: **Reliability caps precision only while nothing can check the descriptor.** ESS of a descriptor against its precision r with n demonstrations in hand (ESS_n of Definition 1), one panel per reliability; $d = 16$, $a_0 = 10$, log scales, one seed of 8,000 prompts. Dotted: the limit of the standalone ESS as $r \rightarrow 0$, 135 demonstrations at $p = 0.99$ and 25 at $p = 0.9$. Alone, the $p = 0.9$ descriptor reaches 23 demonstrations at $r = 0.001$, close to that limit, while the correctly specified one keeps rising; with 100 demonstrations in hand the three panels nearly coincide (109, 107, and 88 at $r = 0.001$).

The identification cost $\mathbb{E}[-\log \pi_n(Z)]$ is the conditional entropy $H(Z | D_n)$. Without demonstrations ($n = 0$), the regret is bounded below by $(1 - p)R^{\text{ex}}(0)$. Consequently, the ESS strictly saturates at a finite value, preventing arbitrary scaling with precision as $r \rightarrow 0$.

Proof idea. Compare the predictor with an oracle twin that is also told Z . The twin’s regret is the left side of (5), and the predictor cannot beat it, since additional information never hurts a Bayes-optimal predictor. Conversely, the predictor’s posterior predictive places weight $\pi_n(Z)$ on the twin’s answer, so it loses at most $-\log \pi_n(Z)$; with no demonstrations that is $\log(1/p)$ when the descriptor is correctly specified and $\log(1/(1 - p))$ when it is misspecified, which average to $H(p)$. More demonstrations can only sharpen the posterior on Z in expectation. The upper bound is the standard bound for a mixture forecaster under log loss, which pays at most the log of one over the weight it placed on the right component (Cesa-Bianchi and Lugosi, 2006, Cor. 3.1 and §9.2) (Merhav and Feder, 1998, §V), and the lower bound is the Bayes-optimality of the informed twin; what is new is what the two bounds imply for the ESS. The proof is in Appendix B.2.

Without demonstrations, reliability caps the ESS. With no demonstrations nothing in the prompt distinguishes the two components, since m has the same law whether or not it is correctly specified, so the predictor must hedge. Proposition 2 caps the ESS at the n where $R^{\text{ex}}(n) = (1 - p)R^{\text{ex}}(0)$: 40 demonstrations at $p = 0.9$ and 325 at $p = 0.99$, for $d = 16$ and $a_0 = 10$. The actual limit is lower, because in the precise limit the identification cost is all of $H(p)$: a descriptor that pins w down exactly or is unrelated to it leaves nothing to tell the two cases apart. As $r \rightarrow 0$ the ESS saturates at 25 demonstrations at $p = 0.9$ and 135 at $p = 0.99$ (dotted in Figure 2). The bounds are close to the computed regret: at $r = 0.001$ and $p = 0.9$ they are 0.32 and 0.64 nats and the regret is 0.57, against 2.51 with the prior alone and 0.07 for a correctly specified descriptor.

Table 3 (Appendix A) shows the saturation across precision. Refining r from 0.1 to 0.001 raises the ESS of a correctly specified descriptor from 14.9 to 116, but that of a descriptor with $p = 0.9$ only from 13.1 to 23.1. A one-in-a-hundred chance of misspecification nearly halves the ESS of the most precise descriptor, 64 against 116, yet leaves coarse descriptors almost untouched: the $p = 0.99$ curve stays within a tenth of the correctly specified one down to $r = 0.01$. The same saturation appears at $d \in \{4, 64\}$ and at $a_0 = 100$.

Demonstrations resolve unidentifiability. As demonstrations accumulate, they provide evidence to infer the latent indicator Z , driving the conditional entropy $H(Z | D_n)$ down. In (5) the lower bound $(1 - p)R^{\text{ex}}(n)$ falls with n and the identification cost cannot rise, so the cap rises as demonstrations accumulate. Once the demonstrations have identified the descriptor and $R^{\text{ex}}(n) \approx d/(2n)$, the correctly specified regret is about $d/(2(n + c))$, and setting $d/(2n^*)$

equal to the lower bound gives

$$\text{ESS}_n \approx \frac{n p c}{n + (1 - p) c} \xrightarrow{n \rightarrow \infty} p c = \frac{p}{r a_0}, \quad (6)$$

against $\text{ESS}_n \rightarrow c$ for a correctly specified descriptor. The ESS of a misspecified descriptor then approaches p times that of a correctly specified one ($\text{ESS}_n \rightarrow p c$). Identification requires the two components to be distinguishable: the identification cost tends to the residual entropy of Z given w and m , which is negligible here because the components are far apart (their Kullback–Leibler divergence is of order $\frac{d}{2} \log(1/r)$) but stays positive for a coarse descriptor, 0.17 nats at $d = 5$, $r = 0.5$.

Before this limit the exact predictor passes through two regimes, shown in Figure 2(c) for $r = 0.001$, $p = 0.9$, $d = 16$, and $c = 100$. The first demonstration or two identify the descriptor: the identification cost falls from $H(p) = 0.33$ nats to 0.07 after one demonstration and 0.02 after two, and the ESS rises from 23 to 30. The ESS then stays near 30 while $n < d$, because the misspecified component still carries the full demonstrations-only regret, and rises once $n > d$ as that regret shrinks: 29 at $n = 10$, 37 at $n = 16$, and 88 at $n = 100$, against 82 from (6) and a limit of 90. Over the same range the correctly specified descriptor’s ESS drifts from 117 to 108, so at $n = 100$ the potentially misspecified descriptor is worth four fifths of the correctly specified one, where alone it was worth a fifth. A descriptor with $c < d$ has little left to verify: at $r = 0.01$, where $c = 10$, the potentially misspecified descriptor gains one demonstration after the first and then loses ESS as the correctly specified one does. Demonstrations effectively pay down the informational cost of potential misspecification.

The cost of overconfidence. The cap and its lifting assume a predictor calibrated to p . An overconfident predictor ($q = 1$) interacting with a misspecified descriptor incurs severe regret. Because the assumed prior is highly concentrated around the erroneous descriptor, the true generative target falls drastically outside the posterior predictive distribution. For highly precise descriptors, overconfidence is an active liability, yielding higher regret than ignoring the descriptor entirely. With reliance $q = 1$ when the truth is $p = 0.9$, the predictor pays for the wrong tenth in full. At $d = 16$ and $a_0 = 10$ the descriptor alone costs it 2.2 nats at $r = 0.1$, against 2.51 for no descriptor, so a coarse descriptor still helps (6.5 demonstrations, against 13.2 when oracle-calibrated). At $r = 0.01$ it costs 6.3 nats and at $r = 0.001$ it costs 13.4, worse than no descriptor at all. Demonstrations repair the damage only slowly, because the wrong tenth carries a prior as tight as the descriptor: after 100 demonstrations the $r = 0.01$ descriptor still leaves the predictor 58 demonstrations behind one without a descriptor, and the $r = 0.001$ descriptor leaves 4.0 nats of regret against 0.09. This is the overconfident predictor of §3; §6 shows that a network trained only on correctly specified descriptors behaves like it. Figure 3 draws the whole range of reliance: the regret is flat for a wide band below p , rises gently above it, and jumps at $q = 1$; with ten demonstrations in hand the band is flat everywhere except the jump, which demonstrations do not remove.

6 Empirical Analysis: Meta-Trained Transformers

We meta-trained 12-layer Transformers on sequences generated by our problem formulation ($d = 5$, $a_0 = 10$). The descriptor m is provided via a dedicated prefix token, optimized over Gaussian log-loss. A network meta-trained on prompts from this process is not the Bayes-optimal predictor: it has an architecture, an output head, and a training distribution, and each can leave a mark. We ask two questions. Does a trained network value a descriptor as the Bayes-optimal predictor does, and where it does not, what determines the difference? And what does a network learn about descriptors it was never shown misspecified?

Setup. We compare the networks with the Bayes-optimal predictor on the same prompts, following Genewein et al. (2025). Prompts use the prefix embedding of Huang and Ge (2025): an optional descriptor token carrying m , up to 20 demonstration tokens each holding an input beside its label, and a final query token, with two indicator coordinates marking the token type. There is no positional encoding, so the demonstrations are exchangeable. The architecture and optimiser follow Garg et al. (2022): 12 layers, 8 heads, width 256, and Adam at a learning rate of 10^{-4} , with fresh prompts at every step. The network outputs a mean and a log variance for the query’s label and is trained on the Gaussian log loss, so its regret is on the scale of (4). With $d = 5$ and $a_0 = 10$ the network sees $w \sim \mathcal{N}(0, I)$ and noise variance 0.1, the same task in the units of Garg et al. (2022). We draw the number of demonstrations uniformly from 0 to 20 and omit the descriptor in half the prompts. We train one network for each combination of a precise ($r = 0.01$) or coarse ($r = 0.1$) descriptor with a correctly specified ($p = 1$) or potentially misspecified ($p = 0.9$) one. The

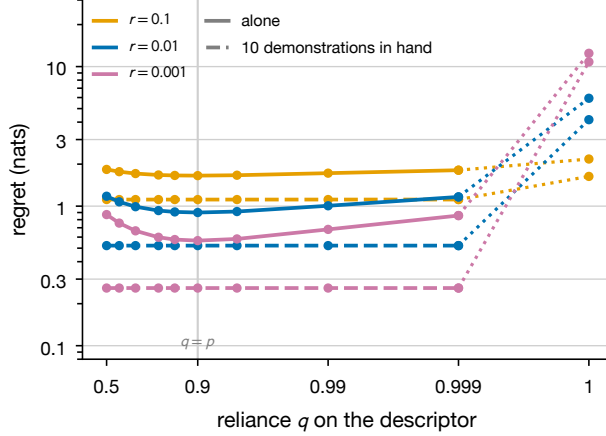


Figure 3: **Under-relying on a descriptor is cheap; relying on it fully is not.** Regret of the Bayes-optimal predictor against the reliance q it places on a descriptor that is correctly specified with probability $p = 0.9$, alone (solid) and with ten demonstrations in hand (dashed), at $d = 16$, $a_0 = 10$; one seed of 8,000 prompts. At $r = 0.001$ alone, reliance 0.85 costs 0.01 nats more than the oracle-calibrated 0.566 and full reliance costs 12.5; with ten demonstrations every reliance below 1 gives 0.26 and full reliance still 10.8.

descriptor token supplies only m , so each network must learn r and p from its meta-training distribution. Each trains for 40,000 steps at batch 1024 without a curriculum (Garg et al. (2022) use batch 64, 500,000 steps, and a curriculum over dimension and prompt length), one training run per setting. In a pilot, the mean regret gap to the Bayes-optimal predictor fell from 0.07 nats at 5,000 steps to 0.03 at 20,000 and stayed near 0.02 from 30,000 to 40,000.

Readouts and reference predictors. We evaluate each network on 20,000 fresh prompts for every n from 0 to 20, with and without a descriptor, and on prompts of both reliabilities, so that each network meets descriptors it was trained on and descriptors it was not. As in Definition 1, the network’s ESS is where its regret with the descriptor alone meets its own regret curve for demonstrations alone. We place each network between the reference predictors of §3 on the same prompts: the *oracle-calibrated* predictor, with reliance equal to the test reliability, and the *training-calibrated* predictor, with reliance equal to the training reliability, which is the predictor each network is trained toward and becomes the overconfident predictor when a network trained at $p = 1$ meets potentially misspecified descriptors. A third reference belongs to the architecture: the *best single Gaussian*, the predictive that matches the mean and variance of the Bayes-optimal mixture, which is all that a mean-and-variance output head can represent. We also read off each network’s implied reliance. The standard error of each regret is below 0.02 nats, except for the precise $p = 1$ model tested at $p = 0.9$, where it reaches 0.10 with a descriptor at small n ; the demonstrations-only curve falls by about 0.13 nats per demonstration near $n = 0$, so these move an ESS by at most 0.15 demonstrations, and by 0.8 in that last case.

Expressivity bounds Bayes-optimal behavior. On their respective training distributions, Transformers match the Bayes-optimal ESS for perfectly specified descriptors. The precise model reaches 14.7 against 15.4 and the coarse one 5.24 against 5.28, with regret 0.02 to 0.03 nats above the Bayes-optimal predictor’s on average over n (Table 1, Figure 4a, b). However, for potentially misspecified descriptors with zero demonstrations, the networks fall short of the exact Bayes-optimal mixture ESS. They reach 4.3 against 6.8 at $r = 0.01$ and 3.6 against 4.2 at $r = 0.1$, with regret 0.5 and 0.2 nats too high, and their implied reliance reads 0.85 rather than 0.9. The shortfall is the output head, not the learning. With a potentially misspecified descriptor and no demonstrations, the Bayes-optimal predictive is two Gaussians, one around the descriptor’s answer and one around the prior’s. Standard regression heads emit a single mean and variance, inherently lacking the capacity to represent a bimodal posterior mixture. Consequently, the networks project the optimal mixture onto the *best single Gaussian* via moment matching. Its ESS is 4.31 at $r = 0.01$ and 3.62 at $r = 0.1$: the networks’ 4.30 and 3.57. As demonstrations collapse the posterior mixture to a single component, the networks successfully converge with the exact Bayes-optimal predictor. By $n = 5$ the best single Gaussian is within 0.01 nats of the mixture and the network within its usual 0.05 (Figure 4c, d), and the network’s ESS with demonstrations in hand rises from 4.3 to 7.3 after five and 8.1 after ten (mixture 8.8 and 9.8), reproducing the lifting of §5; at $r = 0.1$

Table 1: Networks against the reference predictors ($d = 5$, $a_0 = 10$, one training run per model, 20,000 evaluation prompts). The first four rows test each network on its meta-training distribution, the last four on the other reliability. “Trained” is the training-calibrated predictor, the Bayes-optimal predictor with reliance equal to p_{train} ; “calibrated” the oracle-calibrated one with reliance equal to p_{test} (shown where the two differ); and “single Gaussian” the best predictive a mean-and-variance head can express, shown where the Bayes-optimal predictive is a two-component mixture. The last two columns are the mean regret gap, network minus the training-calibrated predictor, over $n = 0, \dots, 20$, with and without a descriptor.

r	p_{train}	p_{test}	ESS				Regret gap (nats)	
			Network	Trained	Calibrated	Single Gaussian	Descriptor	None
0.1	1	1	5.24	5.28	—	—	0.017	0.016
0.1	1	0.9	2.32	2.05	4.26	—	0.010	0.016
0.01	1	1	14.74	15.35	—	—	0.026	0.030
0.01	1	0.9	0.00	≤ 0	6.92	—	-0.186	0.030
0.1	0.9	1	4.63	5.05	5.25	—	0.025	0.018
0.1	0.9	0.9	3.57	4.22	—	3.62	0.034	0.018
0.01	0.9	1	6.39	11.90	15.64	—	0.055	0.022
0.01	0.9	0.9	4.30	6.82	—	4.31	0.073	0.022

it falls, from 3.6 to 2.3 and 1.3, as the mixture’s does. The implied reliance of 0.85 is what a unimodal predictive looks like through posterior means alone, not a belief the network holds about descriptors. Two controls support this reading: the shortfall is absent wherever the Bayes-optimal predictive is a single Gaussian (every correctly specified row, and every n beyond a few demonstrations), and it is not under-training, since a model trained at $r = 0.05$, $p = 0.9$ with the same recipe and continued to 80,000 steps on fresh prompts kept its ESS (3.9 against 5.0). With one training run per setting we state this as what the runs show, not as a general law.

Reliance is inherited from meta-training. We demonstrate that the implied reliance of the network is governed strictly by the meta-training distribution. The last four rows of Table 1 test each network on the reliability it was not trained on. Models trained exclusively on perfect reliability ($p = 1$) learn an implied reliance of $q \approx 1$. At deployment, they act as overconfident predictors, suffering catastrophic regret on precise but misspecified descriptors. When one descriptor in ten is misspecified, the precise model’s regret with the descriptor alone is 2.87 nats against 1.87 without one, so a descriptor hurts it, its ESS is zero where the oracle-calibrated predictor obtains 6.9, and its implied reliance is 0.999; the overconfident Bayes-optimal predictor does worse still, 3.32 nats. The coarse model is forgiven most of its reliance, ESS 2.3 against the oracle-calibrated 4.3, because a misspecified coarse descriptor does little harm. Conversely, models trained on $p = 0.9$ learn to maintain structural uncertainty, exhibiting an implied reliance of $q \approx 0.85$ even when evaluated exclusively on correctly specified descriptors. Their ESS is 6.4 against the oracle-calibrated 15.6 at $r = 0.01$ and 4.6 against 5.3 at $r = 0.1$. Reliance, then, is learned from the meta-training distribution (in the pilot at $p = 1$, the standalone implied reliance rose from 0.9 at 5,000 steps to 0.99 by 10,000 and stayed there) and carried into deployment unchanged, and the exact regret as a function of reliance (Figure 3) says which error is the expensive one: at $d = 5$, $r = 0.01$, $p = 0.9$, reliance 0.85 instead of 0.9 costs 0.006 nats and full reliance costs 2.8. A learner that has never encountered a misspecified descriptor pays the full price of unreliability on precise descriptors; one that has pays a small price for caution wherever they turn out to be correctly specified.

7 Discussion and Limitations

Our theoretical framework establishes that the ESS of a valid task descriptor scales with its precision and constrained dimensionality. However, potential misspecification introduces an identifiability bottleneck, forcing optimal predictors to maintain uncertainty and capping the descriptor’s value. Demonstrations are critical not just as additive data, but as verification signals that resolve this unidentifiability. Meta-trained Transformers approximate this Bayes-optimal reasoning, bounded by the expressivity of their unimodal output representations, with their propensity for reliance statically inherited from training.

Three points follow. First, the hedge is forced only while nothing can check the descriptor: a few demonstrations

identify it, and a precise, potentially misspecified descriptor is then worth more than it was alone, approaching p times a correctly specified one. Reliability is therefore not a fixed discount but a cost that demonstrations pay down. Second, this holds for a predictor calibrated to the descriptor’s reliability. To an overconfident predictor, as a network trained only on correctly specified descriptors becomes, a precise descriptor that is sometimes wrong is worse than none, and demonstrations repair the damage slowly. Third, hedging requires a predictive that can be two things at once: a network whose head emits one mean and one variance is Bayes-optimal within that class and pays for a potentially misspecified descriptor alone with the gap between the best single Gaussian and the mixture, which demonstrations close. A language model’s distribution over tokens can represent such a mixture; a regression head cannot.

Our analysis is subject to limitations, notably the assumption of a linear-Gaussian generative process. The setting was chosen for exact calculation; whether our claims carry over to large language models reading text is open, and the network results rest on one training run per setting. Furthermore, our ESS metric is defined via one-step prediction; under sequential prediction horizons where the predictor’s own feedback accumulates over time, the ESS dynamics shift, representing a valuable direction for future research. Over such horizons a correctly specified descriptor’s ESS tends to the number of demonstrations whose expected information gain matches its $\frac{d}{2} \log(1/r)$ nats, below the one-step value (7.8 against 14.9 at $d = 16$, $a_0 = 10$, $r = 0.1$); the horizon dependence of potentially misspecified descriptors is left for future work.

References

- Ekin Akyürek, Dale Schuurmans, Jacob Andreas, Tengyu Ma, and Denny Zhou. What learning algorithm is in-context learning? Investigations with linear models. In *ICLR*, 2023.
- Tom B. Brown et al. Language models are few-shot learners. In *NeurIPS*, 2020.
- Nicolò Cesa-Bianchi and Gábor Lugosi. *Prediction, Learning, and Games*. Cambridge University Press, 2006.
- Michael Evans and Hadas Moshonov. Checking for prior-data conflict. *Bayesian Analysis*, 1(4):893–914, 2006.
- Shivam Garg, Dimitris Tsipras, Percy Liang, and Gregory Valiant. What can transformers learn in-context? A case study of simple function classes. In *NeurIPS*, 2022.
- Tim Genewein, Li Kevin Wenliang, Jordi Grau-Moya, Anian Ruoss, Laurent Orseau, and Marcus Hutter. Understanding prompt tuning and in-context learning via meta-learning. In *NeurIPS*, 2025.
- Ritwik Gupta, Rodolfo Corona, Jiaxin Ge, Eric Wang, Dan Klein, Trevor Darrell, and David M. Chan. Enough coin flips can make LLMs act Bayesian. In *ACL*, 2025.
- Ukyo Honda, Soichiro Murakami, and Peinan Zhang. Distilling many-shot in-context learning into a cheat sheet. In *Findings of EMNLP*, 2025.
- Ruomin Huang and Rong Ge. Task descriptors help transformers learn linear models in-context. In *ICLR*, 2025.
- Teven Le Scao and Alexander M. Rush. How many data points is a prompt worth? In *NAACL*, 2021.
- Ziqian Lin and Kangwook Lee. Dual operating modes of in-context learning. In *ICML*, 2024.
- Ziqian Lin, Shubham Kumar Bharti, and Kangwook Lee. In-context learning with hypothesis-class guidance. *arXiv:2502.19787*, 2025.
- Neri Merhav and Meir Feder. Universal prediction. *IEEE Transactions on Information Theory*, 44(6):2124–2147, 1998.
- Satoshi Morita, Peter F. Thall, and Peter Müller. Determining the effective sample size of a parametric prior. *Biometrics*, 64(2):595–602, 2008.
- Robb J. Muirhead. *Aspects of Multivariate Statistical Theory*. Wiley, 1982.
- Beat Neuenschwander, Sebastian Weber, Heinz Schmidli, and Anthony O’Hagan. Predictively consistent prior effective sample sizes. *Biometrics*, 76(2):578–587, 2020.

- Madhur Panwar, Kabir Ahuja, and Navin Goyal. In-context learning through the Bayesian prism. In *ICLR*, 2024.
- Allan Raventós, Mansheej Paul, Feng Chen, and Surya Ganguli. Pretraining task diversity and the emergence of non-Bayesian in-context learning for regression. In *NeurIPS*, 2023.
- Matthew Reimherr, Xiao-Li Meng, and Dan L. Nicolae. Prior sample size extensions for assessing prior impact and prior-likelihood discordance. *Journal of the Royal Statistical Society: Series B*, 83(3):413–437, 2021.
- Yuriy A. Reznik. The optimal discounting parameter of the power prior under predictive log-loss. *arXiv:2608.12159*, 2026.
- Heinz Schmidli, Sandro Gsteiger, Satrajit Roychoudhury, Anthony O’Hagan, David Spiegelhalter, and Beat Neuen-schwander. Robust meta-analytic-predictive priors in clinical trials with historical control information. *Biometrics*, 70(4):1023–1032, 2014.
- X. Tong, Y. Zeng, and J. Zhang. Demonstrations, CoT, and prompting: a theoretical analysis of ICL. *arXiv:2603.19611*, 2026.
- Dietrich von Rosen. Moments for the inverted Wishart distribution. *Scandinavian Journal of Statistics*, 15(2):97–109, 1988.
- Albert Webson and Ellie Pavlick. Do prompt-based models really understand the meaning of their prompts? In *NAACL*, 2022.
- Sang Michael Xie, Aditi Raghunathan, Percy Liang, and Tengyu Ma. An explanation of in-context learning as implicit Bayesian inference. In *ICLR*, 2022.
- Qingyang Zhu, Eric Karl Oermann, and Kyunghyun Cho. Multi-task Bayesian in-context learning. In *ICML*, 2026. *arXiv:2606.20538*.

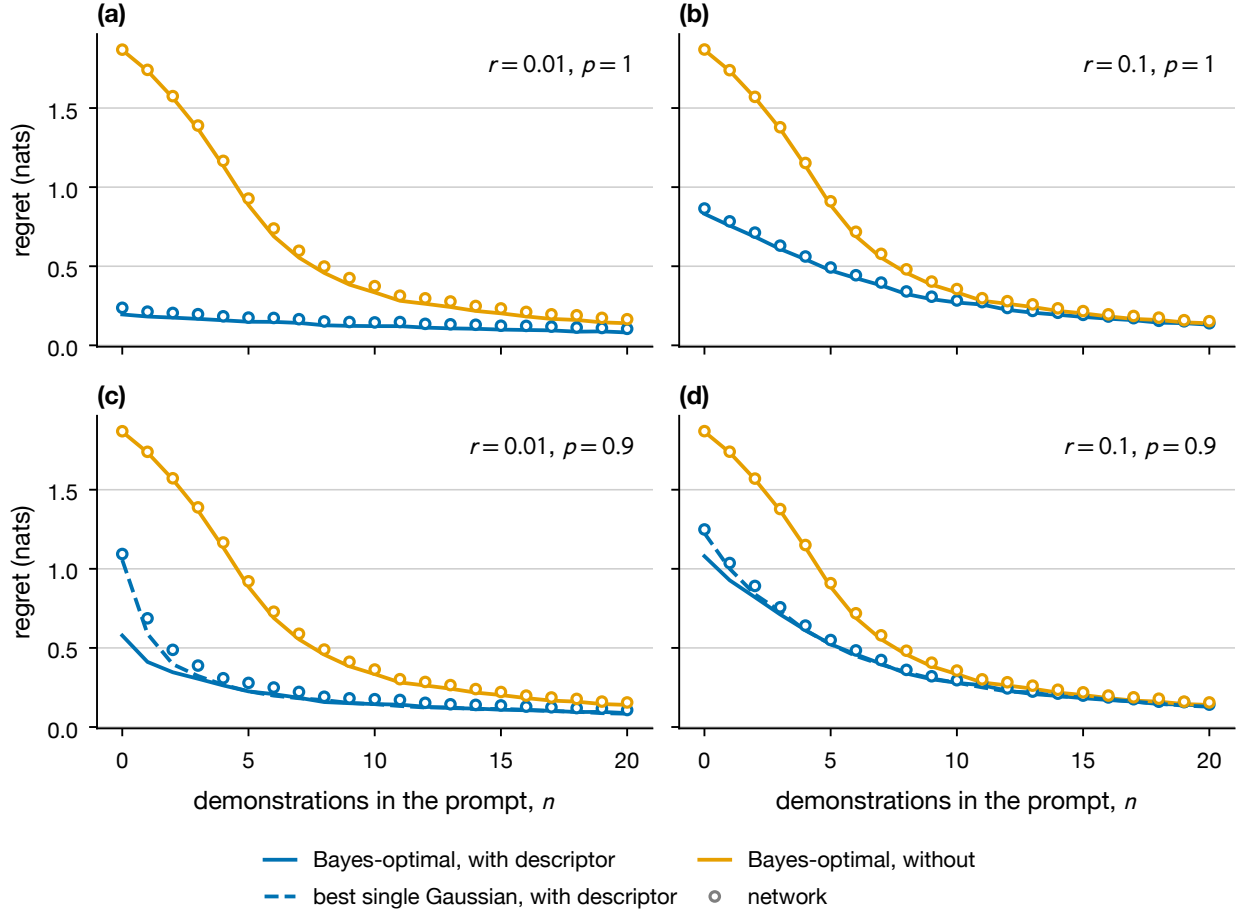


Figure 4: Networks are Bayes-optimal within their output class: they match the mixture where it is a single Gaussian and the best single Gaussian where it is not. Regret against the number of demonstrations for the network (markers; standard errors over 20,000 prompts are smaller than the markers) and the Bayes-optimal predictor (lines), one training run per panel. In each panel the lower curve is prompts with a descriptor followed by n demonstrations and the upper curve is prompts with n demonstrations alone; the ESS is the n at which the upper curve reaches the lower curve's value at $n = 0$. (a) $r = 0.01, p = 1$; (b) $r = 0.1, p = 1$; (c) $r = 0.01, p = 0.9$; (d) $r = 0.1, p = 0.9$. In (c) and (d) the Bayes-optimal predictive with a descriptor is a two-component mixture; the dashed line is the best single Gaussian, and the network lies on it.

A Exact Values

Tables 2 and 3 give the exact one-step ESS behind §4 and §5 at the ends of the grids; Figure 2 shows the relationships between them. The two tables come from separate simulations, so the one quantity they share, the correctly specified descriptor at $d = 16$, $r = 0.001$, differs in its last digit: 117 against 116.

Table 2: Exact ESS of a correctly specified descriptor ($p = 1$) for $a_0 = 10$, three seeds (standard deviation across seeds below 2% of each value), against the rule $(d - 1)(1 - r) + 1/(ra_0)$ of §4, which is within 2.2 demonstrations of every cell.

d	r	ESS	$(d - 1)(1 - r) + 1/(ra_0)$
4	0.1	4.41	3.7
4	0.01	14.5	13
4	0.001	105	103
16	0.1	14.9	14.5
16	0.01	26.1	24.9
16	0.001	117	115
64	0.1	57.7	57.7
64	0.01	73.3	72.4
64	0.001	164	163

Table 3: ESS of potentially misspecified descriptors ($d = 16$, $a_0 = 10$), three seeds; standard deviation across seeds at most 2% of each value. Down a column precision rises tenfold per row; across a row the descriptor is misspecified never, one time in a hundred, and one time in ten.

r	$p = 1$	$p = 0.99$	$p = 0.9$
0.1	14.9	14.6	13.1
0.01	26.1	24.2	18.4
0.001	116	64.2	23.1

B Mathematical Proofs

B.1 Proof of Proposition 1 (Correctly Specified Descriptors)

We prove Proposition 1 using the exact regret expansions for the predictive log-loss. Let ψ denote the digamma function, and let $\varepsilon = 1/a_0$. Given n examples $X_n \in \mathbb{R}^{n \times d}$, the posterior covariance is $\Sigma = \sigma_y^2(\varepsilon I + X_n^\top X_n)^{-1}$. For a query $x \sim \mathcal{N}(0, I_d)$, let $A = \|x\|^2 \sim \chi_d^2$. The one-step expected regret is $R^{\text{ex}}(n) = \frac{1}{2} \mathbb{E} \log(1 + x^\top \Sigma x)$.

Regret Expansions. By leveraging properties of Wishart distributions and the Sherman-Morrison formula, we can bound the regret in two regimes:

1. **Fewer than d examples** ($n < d$): Let $k = d - n$. The regret is lower bounded by:

$$2R^{\text{ex}}(n) \geq \log(2a_0) + \psi(k/2) \quad (7)$$

with equality as $a_0 \rightarrow \infty$.

2. **At least d examples** ($n \geq d$): Let $m = n - d + 1$. The regret is upper bounded by:

$$2R^{\text{ex}}(n) \leq \psi\left(\frac{n+1}{2}\right) - \psi\left(\frac{m}{2}\right) \quad (8)$$

Locating the ESS Root. By Definition 1, the ESS is the root n^* of $R^{\text{ex}}(n^*) = R^{\text{desc}}(0)$. For a correctly specified descriptor, $R^{\text{desc}}(0) = \frac{1}{2}\mathbb{E} \log(1 + c^{-1}A)$ where $c = 1/(ra_0)$. Matching the regret expansions to $R^{\text{desc}}(0)$ and applying the asymptotic expansions of the digamma function ($\psi(z) = \log z - 1/(2z) + O(z^{-2})$) yields the roots:

- In the coarse regime ($c \leq 1$), matching the lower bound gives $d - n \approx rd$, leading to $\text{ESS} \approx (d - 1)(1 - r) + (1 - r)c$.
- In the precise regime ($c \geq d$), matching the upper bound gives $n \approx c + d + 1$, leading to $\text{ESS} \approx c + d + 1 - 1/a_0$.

By concavity of the log-loss, the ESS is strictly bounded across all regimes by $c + d + 2$, concluding the proof.

The remainder of this section makes the sketch rigorous: it states the proposition in a fuller form, with the regret expansions behind (1) and (2) and their error terms, proves the two bounds above as lemmas with their expansions, and locates the root.

Proposition 1 (full statement). *Let $p = 1$, let ψ be the digamma function, and let $c = 1/(ra_0) = \sigma_y^2/s_\ell^2$ with $s_\ell^2 = rs_0^2$: the ESS of the descriptor's prior $\mathcal{N}(m, s_\ell^2 I_d)$ in the conjugate sense, noise variance over prior variance (Neuenschwander et al., 2020), the conjugate-prior ESS.*

- (i) Coarse descriptors. Suppose $c \leq 1$. Then for every $n < d$, $R^{\text{ex}}(n) \geq \frac{1}{2}[\log(2a_0) + \psi(\frac{d-n}{2})]$, with equality in the limit $a_0 \rightarrow \infty$, and as $rd \rightarrow \infty$, uniformly in $c \leq 1$,

$$\text{ESS} = (d - 1)(1 - r) + (1 - r)c + O\left(\frac{1}{rd}\right).$$

- (ii) Precise descriptors. Suppose $c \geq d$. Then for every $n \geq d$, $R^{\text{ex}}(n) \leq \Phi(n) := \frac{1}{2}[\psi(\frac{n+1}{2}) - \psi(\frac{n-d+1}{2})]$, with equality in the limit $a_0 \rightarrow \infty$, and as $c/d \rightarrow \infty$, uniformly in $a_0 \geq 1$,

$$\text{ESS} = c + d + 1 - \frac{1}{a_0} + O\left(\frac{d}{c}\right).$$

For every d, a_0 , and r , $\text{ESS} \leq c + d + 2$.

The two regimes say the same thing. In both regimes the ESS of a correctly specified descriptor is its high-SNR term plus its conjugate-prior ESS, $\text{ESS} \approx (d - 1)(1 - r) + 1/(ra_0)$, to within about two demonstrations. In (i) the conjugate term enters as $(1 - r)c$, which falls short of c by $1/a_0 < 1$; in (ii) the constant $c + d + 1 - 1/a_0$ exceeds the rule by $2 - 1/a_0 + (d - 1)r \in [1, 2)$, up to the stated error. The regimes meet at $c = 1$, where the descriptor constrains w as tightly as one label's noise does, and the ESS passes d near $c = d$. Below that point each gain in precision removes directions; above it every direction has already been probed, and the ESS grows like c , the demonstrations needed to match the descriptor's variance, plus the $d + 1$ that d noisy demonstrations cannot supply. The bound (iii) holds throughout, including the crossover $1 < c < d$ where neither expansion applies. On the grid of Table 2 (Appendix A) the rule of §4 is within 2.2 demonstrations of the exact ESS in every cell with $a_0 \geq 10$; at $d = 64$, $a_0 = 100$, $r = 0.1$ the exact ESS is 56.85 against 56.79 from (i), and at $d = 16$, $a_0 = 10$, $r = 0.001$ it is 116.8 against 116.9 from (ii).

Standard facts. Throughout, $\varepsilon = 1/a_0$, $A = \|x\|^2 \sim \chi_d^2$ for the query x , $\Sigma = (\varepsilon I + X_n^\top X_n)^{-1}$, and $g(t) = \mathbb{E} \log(1 + At)$, which is increasing and concave with

$$d(1 - (d + 2)t) \leq g'(t) = \mathbb{E} \frac{A}{1 + At} \leq d, \quad -(d^2 + 2d) \leq g''(t) \leq 0. \quad (9)$$

We use (a) $\mathbb{E} \log \chi_k^2 = \psi(k/2) + \log 2$, $\mathbb{E}[\chi_k^{-2j}] = \prod_{i=1}^j (k - 2i)^{-1}$ for $k > 2j$, and $\psi(z) = \log z - 1/(2z) + O(z^{-2})$; (b) for $W \sim W_q(p, I)$ with $p \geq q$ and $z \sim \mathcal{N}(0, I_q)$ independent of W , $z^\top W^{-1} z$ is distributed as χ_q^2/χ_{p-q+1}^2 with independent numerator and denominator (Muirhead, 1982, Theorem 3.2.12), and $\mathbb{E} \text{tr} W^{-2} = q(p - 1)/((p - q)(p - q - 1)(p - q - 3))$ for $p - q \geq 4$ (von Rosen, 1988); (c) $y - y^2/2 \leq \log(1 + y) \leq y$ for $y \geq 0$.

Regret. The posterior predictive is the true conditional law of y , a Gaussian with variance $1 + x^\top \Sigma x$, so its expected log loss is its entropy and $R^{\text{ex}}(n) = \frac{1}{2}\mathbb{E} \log(1 + x^\top \Sigma x)$, as in §4. It is strictly decreasing in n , since by the Sherman–Morrison formula each new demonstration lowers $x^\top \Sigma x$ almost surely. For the descriptor, $n = 0$ and $\Sigma = ra_0 I = I/c$, so by (a) and (c) with $y = c/A$,

$$2R^{\text{desc}} = g(1/c) = \log(2ra_0) + \psi(d/2) + \delta^{\text{desc}}, \quad \delta^{\text{desc}} = \mathbb{E} \log(1 + c/A) = \frac{c}{d - 2} + O\left(\frac{c^2}{d^2}\right) \quad (d \geq 5). \quad (10)$$

Locating the root. Let $h(n) = 2R^{\text{ex}}(n) - 2R^{\text{desc}}$ on integers $n \geq 0$, with linear interpolant $\bar{h}$. Then ESS is the unique root of $\bar{h}$, and $h(0) = g(a_0) - g(ra_0) > 0$. In (i) and (ii) we show, for some $n_0 \geq 0$, $\alpha > 0$, and $\theta \rightarrow 0$ along the stated limit,

$$h(n) = \alpha(n_0 - n) + O(\alpha\theta) \quad \text{uniformly over integers } n \geq 0 \text{ with } |n - n_0| \leq 3. \quad (11)$$

Being a convex combination of two such values, $\bar{h}(t)$ obeys the same expansion at real $t \geq 0$ with $|t - n_0| \leq 2$; so for small θ the root lies within 2 of n_0 , and $0 = \bar{h}(\text{ESS}) = \alpha(n_0 - \text{ESS}) + O(\alpha\theta)$ gives $\text{ESS} = n_0 + O(\theta)$.

Lemma B.1 (fewer than d demonstrations). *Let $n < d$ and $k = d - n$. Then $2R^{\text{ex}}(n) = \log(2a_0) + \psi(k/2) + \delta^{\text{ex}}$ with $\delta^{\text{ex}} \geq 0$, $\delta^{\text{ex}} \rightarrow 0$ as $a_0 \rightarrow \infty$, and, for $k \geq 5$,*

$$\delta^{\text{ex}} = \frac{\varepsilon(d-1)}{(k-1)(k-2)} + O\left(\frac{\varepsilon^2 d^2}{k^4}\right).$$

Proof. Write $X_n^\top X_n = \sum_{i \leq n} \lambda_i u_i u_i^\top$ with $\lambda_i > 0$ almost surely, and let Q be the squared norm of the projection of x onto the k directions orthogonal to the u_i . Then Σ has eigenvalue a_0 on those directions and $1/(\varepsilon + \lambda_i)$ on u_i , so

$$1 + x^\top \Sigma x = a_0 Q + T, \quad T = 1 + V, \quad V = \sum_{i \leq n} \frac{z_i^2}{\varepsilon + \lambda_i}, \quad z_i = u_i^\top x,$$

where, given X_n , the z_i are i.i.d. standard normal and independent of $Q \sim \chi_k^2$. Taking logarithms and expectations, by (a),

$$2R^{\text{ex}}(n) = \log(2a_0) + \psi(k/2) + \delta^{\text{ex}}, \quad \delta^{\text{ex}} = \mathbb{E} \log\left(1 + \frac{\varepsilon T}{Q}\right) \geq 0.$$

As $a_0 \rightarrow \infty$, εT decreases to 0 pathwise, so $\delta^{\text{ex}} \rightarrow 0$ by dominated convergence (the integrand at $a_0 = 1$ is integrable).

For the expansion, let $F = \sum_i z_i^2 / \lambda_i$. In the eigenbasis of $W = X_n X_n^\top \sim W_n(d, I)$, which has the same eigenvalues, $F = z^\top W^{-1} z$ and $\sum_i z_i^2 / \lambda_i^2 = z^\top W^{-2} z$ with $z \sim \mathcal{N}(0, I_n)$ independent of W . By (b), $F \sim \chi_n^2 / \chi_{k+1}^2$, so $\mathbb{E} F = n/(k-1)$ and $\mathbb{E}(1+F)^2 = O(d^2/k^2)$, and $\mathbb{E} \sum_i z_i^2 / \lambda_i^2 = n(d-1)/(k(k-1)(k-3))$. Since $1/\lambda - \varepsilon/\lambda^2 \leq 1/(\varepsilon + \lambda) \leq 1/\lambda$, we have $F - \varepsilon \sum_i z_i^2 / \lambda_i^2 \leq V \leq F$. Now (c) with $y = \varepsilon T/Q$, the independence of Q from T , and (a) give

$$\frac{\varepsilon(1 + \mathbb{E} V)}{k-2} - \frac{\varepsilon^2 \mathbb{E} T^2}{2(k-2)(k-4)} \leq \delta^{\text{ex}} \leq \frac{\varepsilon(1 + \mathbb{E} F)}{k-2} = \frac{\varepsilon(d-1)}{(k-1)(k-2)},$$

and the two sides differ by $\varepsilon^2 \mathbb{E} \sum_i z_i^2 / \lambda_i^2 / (k-2) + \varepsilon^2 \mathbb{E} T^2 / (2(k-2)(k-4)) = O(\varepsilon^2 d^2 / k^4)$, as $\mathbb{E} T^2 \leq \mathbb{E}(1+F)^2$. $\square$

Lemma B.2 (at least d demonstrations). *Let $n \geq d$ and $m = n - d + 1$. Then $2\Phi(n) = \mathbb{E} \log(1 + A/Q)$ with $Q \sim \chi_m^2$ independent of A , and $R^{\text{ex}}(n) \leq \Phi(n)$, with equality as $a_0 \rightarrow \infty$. If moreover $a_0 \geq 1$ and $m \geq \max(d, 7)$, then*

$$2R^{\text{ex}}(n) = 2\Phi(n) - \frac{\varepsilon d}{m^2} + O\left(\frac{\varepsilon d^2}{m^3}\right).$$

Proof. Rotate coordinates so that $x = \|x\|e_1$; the rows of X_n remain i.i.d. $\mathcal{N}(0, I_d)$, independent of A . Then $x^\top \Sigma x = A \Sigma_{11} = A/S$ with S the Schur complement of the first coordinate in Σ^{-1} , which the Woodbury identity writes, with X_1 the first column of X_n and X_{-1} the other $d-1$ columns, as

$$S = \varepsilon + X_1^\top (I_n + a_0 X_{-1} X_{-1}^\top)^{-1} X_1.$$

Given X_{-1} , the middle matrix has eigenvalue 1 on the m directions orthogonal to the columns of X_{-1} and $1/(1 + a_0 \mu_j)$ on the other $d-1$, where μ_j are the eigenvalues of $M = X_{-1}^\top X_{-1} \sim W_{d-1}(n, I)$. Since $X_1 \sim \mathcal{N}(0, I_n)$ is independent of X_{-1} , its coordinates in that eigenbasis are i.i.d. standard normal, and

$$S = Q + \Delta, \quad Q \sim \chi_m^2, \quad \Delta = \varepsilon + \sum_{j < d} \frac{z_j^2}{1 + a_0 \mu_j} \in [\varepsilon, \varepsilon(1 + F)], \quad F = \sum_{j < d} \frac{z_j^2}{\mu_j},$$

with A, Q , and (z, M) independent; by (b), $F \sim \chi_{d-1}^2 / \chi_{m+1}^2$, so $\mathbb{E} F = (d-1)/(m-1)$ and, when $m \geq \max(d, 7)$, $\mathbb{E}(1+F)^2 \leq 5$.

Since $A + Q \sim \chi_{n+1}^2$, (a) gives $\mathbb{E} \log(1 + A/Q) = \psi(\frac{n+1}{2}) - \psi(\frac{m}{2}) = 2\Phi(n)$. As $S \geq Q$, $R^{\text{ex}}(n) \leq \Phi(n)$, with equality as $a_0 \rightarrow \infty$ by monotone convergence, since Δ decreases to 0.

For the expansion, $2R^{\text{ex}}(n) = 2\Phi(n) - \mathbb{E}D$ with

$$D = \log\left(1 + \frac{A}{Q}\right) - \log\left(1 + \frac{A}{S}\right) = \log(1 + y_1) - \log(1 + y_2), \quad y_1 = \frac{\Delta}{Q} \geq y_2 = \frac{\Delta}{Q+A}, \quad y_1 - y_2 = \frac{\Delta A}{Q(Q+A)}.$$

By concavity of $\log(1 + y)$ and $1/(1 + y_1) \geq 1 - y_1$, $(y_1 - y_2)(1 - \Delta/Q) \leq D \leq y_1 - y_2$. Here Δ is independent of (A, Q) ; $\mathbb{E} \frac{A}{Q(Q+A)} = \mathbb{E} \frac{1}{Q} - \mathbb{E} \frac{1}{Q+A} = \frac{d}{(m-2)(m+d-2)}$ by (a), as $Q + A \sim \chi_{m+d}^2$; $\frac{A}{Q^2(Q+A)} \leq \frac{A}{Q^3}$; and $\varepsilon \leq \mathbb{E}\Delta \leq \varepsilon(1 + \mathbb{E}F) = \varepsilon \frac{m+d-2}{m-1}$, $\mathbb{E}\Delta^2 \leq 5\varepsilon^2$. Hence

$$\frac{\varepsilon d}{(m-2)(m+d-2)} - \frac{5\varepsilon^2 d}{(m-2)(m-4)(m-6)} \leq \mathbb{E}D \leq \frac{\varepsilon d}{(m-1)(m-2)},$$

and both sides are $\varepsilon d/m^2 + O(\varepsilon d^2/m^3)$, using $\varepsilon \leq 1$. $\square$

Proof of Proposition 1. (iii). Conditionally on A , $\log(1 + Av)$ is concave in v , so Jensen's inequality with $\mathbb{E}[1/Q] = 1/(m-2)$ gives $2\Phi(n) \leq g(1/(m-2))$ for $m \geq 3$. If $m-2 \geq c$, that is $n \geq c + d + 1$, then $g(1/(m-2)) \leq g(1/c) = 2R^{\text{desc}}$, so $R^{\text{ex}}(n) \leq \Phi(n) \leq R^{\text{desc}}$ by Lemma B.2. The least such integer is $\lceil c + d + 1 \rceil < c + d + 2$, and R^{ex} is decreasing, so $\text{ESS} \leq c + d + 2$.

(i). The lower bound and the limit are Lemma B.1. For the expansion let $c \leq 1$, $n_0 = (d-1)(1-r) + (1-r)c$, and $n \geq 0$ an integer with $|n - n_0| \leq 3$; write $k = d - n = rd + u$. Since $n_0 = d - rd - u_0$ with $u_0 = 1 - r + \varepsilon - c \in [-c, 1]$, we have $|u| \leq 4$, so $k \geq 5$ once $rd \geq 9$. By Lemma B.1, (10), $\psi(z) = \log z - 1/(2z) + O(z^{-2})$, and $\varepsilon = rc$, with every O uniform in $c \leq 1$ and $|u| \leq 4$,

$$h(n) = \psi(k/2) - \psi(d/2) - \log r + \delta^{\text{ex}} - \delta^{\text{desc}} \\ = \left[\log \frac{k}{rd} - \frac{1}{k} + \frac{1}{d} \right] + \frac{\varepsilon d}{r^2 d^2} \left(1 + O\left(\frac{1}{rd}\right) \right) - \frac{c}{d} + O\left(\frac{1}{(rd)^2}\right) = \frac{u - 1 + r + c - \varepsilon}{rd} + O\left(\frac{1}{(rd)^2}\right),$$

since $\log(k/(rd)) = u/(rd) + O((rd)^{-2})$, $1/k = 1/(rd) + O((rd)^{-2})$, $\varepsilon d/(r^2 d^2) = c/(rd)$, and $c/d = \varepsilon/(rd)$. So $h(n) = \frac{1}{rd} [(n_0 - n) + O(1/(rd))]$, which is (11) with $\alpha = \theta = 1/(rd)$, and $\text{ESS} = (d-1)(1-r) + (1-r)c + O(1/(rd))$.

(ii). The upper bound and the limit are Lemma B.2. For the expansion let $n_0 = c + d + 1 - \varepsilon$ and n an integer with $|n - n_0| \leq 3$, so that $m - 2 = c + s$ with $s = n - n_0 - \varepsilon$, $|s| \leq 4$; for $c \geq 9d$ every such n has $m \geq \max(d, 7)$. By (10) and Lemma B.2, $2\Phi(n) - 2R^{\text{desc}} = \mathbb{E}[g(1/Q) - g(1/c)]$, and a second-order Taylor expansion of g about $1/c$ with (9), $\mathbb{E}[1/Q] - 1/c = -s/(c(c+s))$, and $\mathbb{E}(1/Q - 1/c)^2 = \text{Var}(1/Q) + s^2/(c(c+s))^2 = O(c^{-3})$ gives

$$2\Phi(n) - 2R^{\text{desc}} = -\frac{s}{c(c+s)} d \left(1 + O\left(\frac{d}{c}\right) \right) + O\left(\frac{d^2}{c^3}\right) = -\frac{ds}{c^2} + O\left(\frac{d^2}{c^3}\right).$$

With Lemma B.2, $\varepsilon \leq 1$, and $\varepsilon d/m^2 = \varepsilon d/c^2 + O(d/c^3)$,

$$h(n) = -\frac{d}{c^2} \left[s + \varepsilon + O\left(\frac{d}{c}\right) \right] = \frac{d}{c^2} \left[(n_0 - n) + O\left(\frac{d}{c}\right) \right],$$

which is (11) with $\alpha = d/c^2$ and $\theta = d/c$, uniformly in $a_0 \geq 1$, so $\text{ESS} = c + d + 1 - 1/a_0 + O(d/c)$. $\square$

B.2 Proof of Proposition 2 (Potentially Misspecified Descriptors)

Let $D_n = (x_{1:n}, y_{1:n}, m, x)$ denote the prompt and the query input. Let $Z \in \{0, 1\}$ be the true validity indicator of the descriptor.

Step 1: The Mixture Predictive. The Bayes-optimal posterior predictive is a π_n -mixture of the correctly specified predictive f_1 and the base prior predictive f_0 , where $\pi_n = P(Z = 1 \mid D_n)$. That is, $q(y) = \pi_n f_1(y) + (1 - \pi_n) f_0(y)$, where f_1 and f_0 are the predictives of the two components given D_n : f_1 is the correctly specified predictor's, and f_0 is the demonstrations-only predictor's, because under $Z = 0$ the descriptor is independent of w and carries no information about y . At $n = 0$, $\pi_0 = p$: seeing m says nothing about Z , because m has the same law $\mathcal{N}(0, (1-r)a_0 I)$ either way, and x is independent of Z , so there is nothing to update on. The components are then $f_1 = \mathcal{N}(m^\top x, 1 + ra_0 \|x\|^2)$ and $f_0 = \mathcal{N}(0, 1 + a_0 \|x\|^2)$.

Step 2: The Oracle Twin Lower Bound. Consider an oracle predictor that is explicitly given Z . This twin predicts using f_1 when $Z = 1$ (incurring regret $R_{p=1}^{\text{desc}}(n)$) and f_0 when $Z = 0$ (incurring regret $R^{\text{ex}}(n)$). The twin's expected regret is exactly $pR_{p=1}^{\text{desc}}(n) + (1-p)R^{\text{ex}}(n)$. Because additional information cannot hurt a Bayes-optimal predictor, the expected log-loss of our mixture predictor q cannot be lower than the oracle twin's log-loss. Explicitly, given (D_n, Z) the true conditional density of y is f_Z , and the expected log loss of any density q exceeds that of f_Z by $\mathbb{E}[-\log q \mid D_n, Z] - \mathbb{E}[-\log f_Z \mid D_n, Z] = \text{KL}(f_Z \parallel q) \geq 0$; averaging over Z and D_n gives the bound. Thus:

$$R^{\text{desc}}(n) \geq pR_{p=1}^{\text{desc}}(n) + (1-p)R^{\text{ex}}(n) \quad (12)$$

Step 3: The Identification Cost Upper Bound. Pointwise, the mixture density satisfies $q \geq \pi_n f_1$ and $q \geq (1-\pi_n)f_0$. Consequently:

$$-\log q \leq -\log f_1 - \log \pi_n \quad (13)$$

$$-\log q \leq -\log f_0 - \log(1 - \pi_n) \quad (14)$$

Taking the expectation over the true data generating process f_Z and the latent variable Z yields the upper bound. The penalty term simplifies to the conditional entropy $\mathbb{E}[-\log \pi_n(Z)] = H(Z \mid D_n)$. At $n = 0$ it is $p \log(1/p) + (1-p) \log(1/(1-p)) = H(p)$.

Step 4: Monotonicity and Saturation. The conditional entropy $H(Z \mid D_n)$ is non-increasing in n , since conditioning on additional demonstrations can only reduce entropy. At $n = 0$, $H(Z \mid D_0) = H(p)$. In detail, the query x is independent of Z and of everything else in D_n , so $H(Z \mid D_n) = H(Z \mid x_{1:n}, y_{1:n}, m)$, and these conditioning sets are nested in n . Crucially, evaluating the lower bound at $n = 0$ gives $R^{\text{desc}}(0) \geq (1-p)R^{\text{ex}}(0)$. Since $R^{\text{ex}}(n)$ is strictly decreasing in n , there exists a finite n^* such that $R^{\text{ex}}(n^*) = (1-p)R^{\text{ex}}(0)$. The ESS cannot exceed this n^* , forcing the strict saturation bottleneck. The lower bound at $n = 0$ holds for every r because $R_{p=1}^{\text{desc}}(0) = \frac{1}{2} \mathbb{E} \log(1 + r a_0 \|x\|^2)$ is nonnegative; and $R^{\text{ex}}(n)$ is non-increasing because a predictor given $n+1$ demonstrations can ignore one, and Bayes-optimal regret cannot rise with more information. Finally $R_{p=1}^{\text{desc}}(0)$ decreases to 0 as $r \rightarrow 0$ by monotone convergence, which gives the two limits of the standalone regret, $(1-p)R^{\text{ex}}(0)$ and $(1-p)R^{\text{ex}}(0) + H(p)$.